

TE 456 - Satellite Communication & NTN

100 Multiple-Choice Questions with Answers

1. A Non-Terrestrial Network (NTN) is best defined as a wireless communication network that:

A. Uses radio network equipment carried on airborne or spaceborne vehicles ✓

B. Relies exclusively on ground-based cell towers and fibre

C. Connects two mobile handsets without any base station

D. Operates only at optical wavelengths between ground stations

Answer: A. Uses radio network equipment carried on airborne or spaceborne vehicles

2. Terrestrial networks (TN) and non-terrestrial networks (NTN) differ primarily in their:

A. Infrastructure and coverage areas ✓

B. Choice of programming language for the core network

C. Use of the TCP/IP protocol suite

D. Billing and subscription models

Answer: A. Infrastructure and coverage areas

3. In NTN terminology, a 'spaceborne' platform refers to:

A. Drones and balloons flying in the lower atmosphere

B. Fibre-optic backbone links between data centres

C. Satellite communication using artificial satellites orbiting the Earth ✓

D. Ground gateways that steer antennas toward satellites

Answer: C. Satellite communication using artificial satellites orbiting the Earth

4. Which of the following are examples of airborne NTN platforms?

A. GEO, MEO and LEO satellites

B. Fixed gNBs and fibre cables

C. Undersea cables and microwave relay towers

D. Drones (UAVs) and High-Altitude Platform Stations (HAPS) ✓

Answer: D. Drones (UAVs) and High-Altitude Platform Stations (HAPS)

5. A High-Altitude Platform Station (HAPS), such as a balloon or airship in the stratosphere, is best characterised as an NTN platform that is:

A. A satellite in medium Earth orbit at about 20,000 km

B. Unmanned, with lower latency than LEO and easier maintainability than a satellite ✓

C. A ground-based mast forming part of the terrestrial network

D. A geostationary spacecraft fixed above the equator

Answer: B. Unmanned, with lower latency than LEO and easier maintainability than a satellite

6. One motivation for NTN cited in the 6G vision is that terrestrial networks currently:

A. Cover less than 40% of the Earth's surface ✓

B. Cover more than 95% of the Earth's surface

C. Are completely immune to natural disasters

D. Provide uniform, seamless coverage over the oceans

Answer: A. Cover less than 40% of the Earth's surface

7. An NTN platform creating large beams to illuminate remote farms, oceans and IoT devices anywhere on Earth illustrates which NTN use-case category?

A. Service continuity

B. Service scalability

C. Service ubiquity ✓

D. Service localization

Answer: C. Service ubiquity

8. A cellular subscriber who boards a cruise ship or an aircraft and would otherwise lose coverage is best served by which NTN use-case category?

- A. Service ubiquity
- B. Service scalability
- C. Service redundancy
- D. Service continuity ✓**

Answer: D. Service continuity

9. Broadcasting the same content, such as a software update or a live event, to many users over a large area with one wide NTN beam illustrates:

- A. Service scalability ✓**
- B. Service continuity
- C. Service ubiquity
- D. Service isolation

Answer: A. Service scalability

10. Deploying terrestrial base stations across sparsely populated deserts, mountains and oceans is generally avoided mainly because:

- A. Radio waves cannot physically travel over open water
- B. Spectrum is legally unavailable outside cities
- C. Satellites legally prohibit ground towers in those areas
- D. It is not cost-effective and often lacks a viable business case ✓**

Answer: D. It is not cost-effective and often lacks a viable business case

11. A major driver of the decades-long success of terrestrial mobile networks has been:

- A. The complete absence of regulation in the sector
- B. International standardization ensuring vendor compatibility and lower costs ✓**
- C. Exclusive reliance on proprietary, closed hardware
- D. The use of a single global operator worldwide

Answer: B. International standardization ensuring vendor compatibility and lower costs

12. Historically, the satellite communications market has been fragmented largely because of:

- A. Excessive standardization that stifled all innovation
- B. Government bans on manufacturing satellites
- C. An unlimited supply of cheap, interchangeable devices
- D. A lack of standardization, making interoperability between vendors difficult ✓**

Answer: D. A lack of standardization, making interoperability between vendors difficult

13. In a TN-versus-NTN comparison, the typical cell size differs in that:

- A. TN cells are massive while NTN cells are only a few metres across
- B. TN cells are small (giving high capacity per cell), whereas NTN cells are massive, tens to thousands of km across ✓**
- C. Both TN and NTN use cells of an identical fixed diameter
- D. NTN cells are always smaller than TN cells to limit interference

Answer: B. TN cells are small (giving high capacity per cell), whereas NTN cells are massive, tens to thousands of km across

14. The infrastructure of a terrestrial network is characterised by:

- A. Satellites, HAPS and drones operating in the sky
- B. Only end-user handsets communicating peer-to-peer
- C. Fixed ground-based base stations (e.g., gNodeB), cables and switching centres ✓**
- D. Optical inter-satellite links operating in orbit

Answer: C. Fixed ground-based base stations (e.g., gNodeB), cables and switching centres

15. The current relationship between the satellite and terrestrial mobile communication industries is best described as:

A. Progressively converging toward TN-NTN integration ✓

- B. Diverging into two fully separate ecosystems
- C. Remaining two permanently isolated industrial chains
- D. Being replaced entirely by wired networks

Answer: A. Progressively converging toward TN-NTN integration

16. The 3rd Generation Partnership Project (3GPP) is significant to NTN because it:

- A. Manufactures and launches communication satellites
- B. Completed the first global 5G NR standard and is extending it to support satellites ✓**
- C. Owns and operates the Starlink constellation
- D. Issues spectrum licences to operators in each country

Answer: B. Completed the first global 5G NR standard and is extending it to support satellites

17. In 5G rollout, NTN is expected to serve unserved areas such as isolated regions, aircraft and vessels, and additionally to serve:

- A. Underserved areas such as sub-urban/rural regions, upgrading limited terrestrial performance ✓**
- B. Only dense urban centres that already have 5G
- C. Only areas that already have a fibre backhaul
- D. Only military installations, never civilian users

Answer: A. Underserved areas such as sub-urban/rural regions, upgrading limited terrestrial performance

18. Generally, compared with NTN, terrestrial networks are described as providing:

- A. Lower reliability and higher bit error rates
- B. Identical reliability under all conditions
- C. Higher reliability and lower bit error rates ✓**
- D. Reliable service only over the open oceans

Answer: C. Higher reliability and lower bit error rates

19. A frequently cited advantage of NTN over TN during earthquakes, floods or armed conflict is that NTN:

- A. Can maintain coverage when terrestrial infrastructure is damaged or destroyed ✓**
- B. Is completely unaffected by weather of any kind
- C. Requires no ground gateways whatsoever
- D. Eliminates the need for any radio spectrum

Answer: A. Can maintain coverage when terrestrial infrastructure is damaged or destroyed

20. Comparing latency, terrestrial networks are typically ultra-low (under about 1 ms), whereas non-terrestrial networks range from:

- A. About 20 ms for LEO up to 500 ms or more for GEO, owing to the large distances ✓**
- B. About 1 ms for LEO up to 5 ms for GEO
- C. A fixed 100 ms regardless of the orbit used
- D. Lower latency than terrestrial networks in every case

Answer: A. About 20 ms for LEO up to 500 ms or more for GEO, owing to the large distances

21. Which statement best summarises the primary limitations of terrestrial networks?

- A. Unlimited reach but uniformly very high latency
- B. Restricted global reach, vulnerability to physical damage, and high cost in rugged terrain ✓**
- C. Perfect disaster resilience but poor data speed
- D. Global ocean coverage but no urban coverage

Answer: B. Restricted global reach, vulnerability to physical damage, and high cost in rugged terrain

22. Regarding mobility support, a key contrast between the two networks is that:

- A. A TN requires frequent handovers, while an NTN can offer seamless mobility across wide areas ✓**

- B. A TN offers seamless wide-area mobility, while an NTN needs constant handovers
- C. Neither a TN nor an NTN supports user mobility at all
- D. Both require a handover only once per day

Answer: A. A TN requires frequent handovers, while an NTN can offer seamless mobility across wide areas

23. Relative to extensive terrestrial deployments in remote regions, NTNs are described as:

- A. Always more expensive, with no coverage benefit
- B. Equivalent in cost but lower in coverage
- C. Potentially more cost-effective, while extending coverage and providing a backup ✓**
- D. Cheaper only because they need no spectrum

Answer: C. Potentially more cost-effective, while extending coverage and providing a backup

24. In the Ghanaian context discussed, rural regions have largely lagged in 4G coverage, leaving many users dependent on:

- A. 5G standalone networks
- B. Satellite broadband that is already deployed
- C. Fibre-to-the-home connections
- D. 3G networks ✓**

Answer: D. 3G networks

25. Compared with a terrestrial network's standard path loss, the signal constraints of an NTN are dominated by:

- A. Negligible path loss and no Doppler effect whatsoever
- B. Interference only from other ground-based cell towers
- C. A complete absence of any propagation impairments
- D. Severe path loss and high Doppler shifts from satellite mobility, plus atmospheric/ionospheric interference ✓**

Answer: D. Severe path loss and high Doppler shifts from satellite mobility, plus atmospheric/ionospheric interference

26. According to Kepler's laws as applied to satellites, satellite orbits are:

- A. Perfect circles centred exactly on the Earth's core
- B. Straight lines tangent to the equator
- C. Squares traced around the two poles
- D. Ellipses with the Earth at one of the foci ✓**

Answer: D. Ellipses with the Earth at one of the foci

27. In an altitude-versus-orbital-speed table, orbital speed falls from about 7.8 km/s near 200 km to about 3.1 km/s at roughly 35,800 km. This illustrates that:

- A. Orbital speed increases as altitude increases
- B. Orbital speed decreases as altitude increases, while the orbital period lengthens ✓**
- C. Orbital speed is the same at every altitude
- D. The orbital period shortens as altitude increases

Answer: B. Orbital speed decreases as altitude increases, while the orbital period lengthens

28. A geostationary (GEO) satellite orbits at an altitude of approximately:

- A. 35,786 km ✓**
- B. 550 km
- C. 20,200 km
- D. 384,000 km

Answer: A. 35,786 km

29. A GEO satellite appears stationary to a fixed observer on Earth because it:

- A. Does not actually move at all in its orbit
- B. Completes one orbit in about 24 hours, matching the Earth's rotation ✓**
- C. Orbits the Earth in under 90 minutes

D. Is physically held in place by a tether

Answer: B. Completes one orbit in about 24 hours, matching the Earth's rotation

30. Theoretically, near-global coverage (excluding the extreme poles) can be achieved with how many GEO satellites?

- A. One
- B. Twelve
- C. Twenty-four
- D. Three ✓**

Answer: D. Three

31. Because of its high altitude, a GEO satellite exhibits a round-trip propagation delay on the order of:

- A. About 1 ms
- B. About 600 ms ✓**
- C. About 25 ms
- D. About 50 microseconds

Answer: B. About 600 ms

32. A recognised disadvantage of GEO satellites is:

- A. The need for constant satellite-to-satellite handovers
- B. An extremely short, negligible propagation delay
- C. A requirement for thousands of satellites for coverage
- D. Poor signal coverage at the extreme polar regions ✓**

Answer: D. Poor signal coverage at the extreme polar regions

33. A satellite whose orbital period matches the Earth's rotation but which may be inclined or elliptical, lacking the strict 0-degree equatorial, fixed-point constraint of a GEO, is classified as:

- A. Low Earth orbit (LEO)
- B. Medium Earth orbit (MEO)
- C. Geosynchronous orbit (GSO), of which GEO is a special case ✓**
- D. A High-Altitude Platform Station (HAPS)

Answer: C. Geosynchronous orbit (GSO), of which GEO is a special case

34. Low Earth Orbit (LEO) satellites typically operate at altitudes of:

- A. 300 to 1,500 km ✓**
- B. 7,000 to 25,000 km
- C. Exactly 35,786 km
- D. Above 100,000 km

Answer: A. 300 to 1,500 km

35. A typical LEO satellite has an orbital period of about:

- A. 24 hours
- B. 6 to 12 hours
- C. 90 minutes ✓**
- D. One week

Answer: C. 90 minutes

36. Because of their low altitude, LEO satellites offer a small one-way propagation delay on the order of:

- A. About 600 ms
- B. About 83 ms
- C. Several seconds
- D. 1 to 5 ms ✓**

Answer: D. 1 to 5 ms

37. Medium Earth Orbit (MEO) satellites orbit at altitudes ranging from about:

- A. 300 to 1,500 km
- B. 7,000 to 25,000 km ✓**
- C. 35,786 km, fixed
- D. Below 300 km

Answer: B. 7,000 to 25,000 km

38. MEO satellites are described as the backbone of global navigation; for example, GPS uses at least:

- A. 24 MEO satellites ✓**
- B. 3 GEO satellites
- C. Thousands of LEO satellites
- D. A single HAPS platform

Answer: A. 24 MEO satellites

39. A navigation satellite orbiting at an altitude of about 20,200 km is described as semi-synchronous, meaning its orbital period is approximately:

- A. 24 hours
- B. 12 hours ✓**
- C. 90 minutes
- D. 1 hour

Answer: B. 12 hours

40. Starlink provides broadband using thousands of LEO satellites flying at about 550 km, achieving a round-trip delay of roughly:

- A. 600 ms
- B. 250 ms
- C. 25 ms ✓**
- D. 2 ms

Answer: C. 25 ms

41. A disadvantage of N GEO (LEO/MEO) systems compared with GEO is that they:

- A. Have far higher latency than GEO satellites
- B. Require dynamic hand-offs and active ground-antenna tracking of moving satellites ✓**
- C. Cannot provide any coverage of polar regions
- D. Need only a stationary dish with no tracking

Answer: B. Require dynamic hand-offs and active ground-antenna tracking of moving satellites

42. For High-Altitude Platform Stations (HAPS), 3GPP focuses on altitudes between:

- A. 300 and 1,500 km
- B. 8 and 50 km ✓**
- C. 35,000 and 36,000 km
- D. 1 and 5 km

Answer: B. 8 and 50 km

43. Because a HAPS maintains its position relative to the Earth's surface, it:

- A. Uses only Earth-moving beams like a fast LEO satellite
- B. Has a round-trip delay of about 600 ms
- C. Uses Earth-fixed beams like a GEO satellite, with delay comparable to terrestrial networks ✓**
- D. Covers a smaller area than a single terrestrial gNB

Answer: C. Uses Earth-fixed beams like a GEO satellite, with delay comparable to terrestrial networks

44. Compared with spaceborne NTN, airborne NTN platforms:

- A. Have a much larger propagation delay than GEO
- B. Are completely immune to strong winds and storms

C. Provide permanent global coverage from a single unit

D. Deploy quickly at lower cost with smaller delay, but face stabilization and weather challenges ✓

Answer: D. Deploy quickly at lower cost with smaller delay, but face stabilization and weather challenges

45. For an Unmanned Aircraft System (UAS) used as an airborne NTN platform, the drone's operating altitude is typically limited to about:

A. 8 to 50 km, up in the stratosphere

B. 300 to 1,500 km, like a LEO satellite

C. 90 to 150 m, depending on country and region ✓

D. 35,786 km, like a GEO satellite

Answer: C. 90 to 150 m, depending on country and region

46. The three types of beams used in an NTN to provide radio coverage are Earth-fixed, Earth-moving, and:

A. Polar-locked beams

B. Quasi-Earth-fixed beams ✓

C. Counter-rotating beams

D. Ground-anchored beams

Answer: B. Quasi-Earth-fixed beams

47. An Earth-fixed beam is one that:

A. Sweeps across a new area at every instant

B. Covers the same fixed geographic region on Earth at all times ✓

C. Covers area X then abruptly jumps to area Y

D. Can only be produced by a LEO satellite

Answer: B. Covers the same fixed geographic region on Earth at all times

48. An Earth-moving beam is characterised by the fact that it:

A. Always illuminates the exact same area permanently

B. Covers a slightly different geographic area at each successive instant as the platform moves ✓

C. Is produced only by stationary GEO satellites

D. Requires beam steering to remain fixed on one spot

Answer: B. Covers a slightly different geographic area at each successive instant as the platform moves

49. With a quasi-Earth-fixed beam covering area X from t1 to t2 and then a different area, continuity over area X is maintained because:

A. The satellite physically stops moving in its orbit

B. One beam covers all geographic areas simultaneously

C. The user device switches to a wired connection

D. An incoming quasi-Earth-fixed beam replaces the outgoing beam over the same area ✓

Answer: D. An incoming quasi-Earth-fixed beam replaces the outgoing beam over the same area

50. GEO satellites, with beam footprints from roughly 200 to 3,500 km, are most commonly used for services such as:

A. Ultra-low-latency competitive online gaming

B. Sub-millisecond industrial control loops

C. Short-range indoor positioning

D. Weather monitoring, TV broadcasting, and remote sensing/positioning ✓

Answer: D. Weather monitoring, TV broadcasting, and remote sensing/positioning

51. In 5G NR-NTN, the 'service link' is:

A. The connection between the satellite and the gateway

B. The direct link between two satellites in orbit

C. The 5G radio access (Uu interface) between the UE and the satellite ✓

D. The wired link between the gateway and the 5GC

Answer: C. The 5G radio access (Uu interface) between the UE and the satellite

52. The 'feeder link' (satellite radio interface, SRI) describes the connection between:

A. The UE and the satellite

B. The gNB and the 5G core network

C. The satellite and the ground gateway ✓

D. Two user devices directly

Answer: C. The satellite and the ground gateway

53. An NTN round-trip time is expressed as $RTT = 2(T_{prop1} + T_{prop2}) + 2 T_{slot} + T_{proc1} + T_{proc2}$. In this expression, the propagation terms T_{prop1} and T_{prop2} correspond to the:

A. UE-to-satellite link and the satellite-to-gateway link ✓

B. Uplink and downlink of a single terrestrial cell

C. Two processing steps carried out inside the gNB

D. Inter-satellite link and the terrestrial fibre backhaul

Answer: A. UE-to-satellite link and the satellite-to-gateway link

54. A transparent (bent-pipe) NTN payload:

A. Amplifies, filters and frequency-translates the signal without onboard baseband processing ✓

B. Fully decodes and re-encodes the signal in orbit

C. Hosts the complete gNB onboard the satellite

D. Terminates the Uu interface on the satellite

Answer: A. Amplifies, filters and frequency-translates the signal without onboard baseband processing

55. With a transparent payload, the NTN gNB is located:

A. Onboard the satellite itself

B. Inside the UE handset

C. Distributed across several satellites

D. On the ground, typically co-located with the gateway ✓

Answer: D. On the ground, typically co-located with the gateway

56. In the transparent-payload architecture, the 5G Uu radio interface terminates at:

A. The satellite payload

B. The terrestrial gNB, not at the satellite ✓

C. The UE only

D. The 5G core network

Answer: B. The terrestrial gNB, not at the satellite

57. A regenerative (decode-and-forward) payload differs from a transparent one in that it:

A. Only amplifies and forwards, with no onboard processing

B. Removes the need for any gateway or 5G core

C. Incorporates gNB functions onboard, terminating the Uu interface at the satellite ✓

D. Cannot support inter-satellite links

Answer: C. Incorporates gNB functions onboard, terminating the Uu interface at the satellite

58. A key latency advantage of the regenerative architecture is that:

A. The satellite hardware becomes far simpler and cheaper

B. It eliminates the need for a service link entirely

C. The single-way latency includes only the service link, reducing round-trip time ✓

D. The Uu interface terminates on the ground

Answer: C. The single-way latency includes only the service link, reducing round-trip time

59. An Inter-Satellite Link (ISL) in a regenerative NTN:

- A. Connects a satellite to a terrestrial fibre ring
- B. Connects NTN payloads to one another, allowing relay toward a distant gateway ✓**
- C. Links a UE directly to the 5G core network
- D. Replaces the feeder link with a wired cable

Answer: B. Connects NTN payloads to one another, allowing relay toward a distant gateway

60. A disaggregated gNB is split into a:

- A. Uplink Unit and a Downlink Unit
- B. Service Unit and a Feeder Unit
- C. Primary Unit and a Secondary Unit
- D. Centralized Unit (CU) and a Distributed Unit (DU) ✓**

Answer: D. Centralized Unit (CU) and a Distributed Unit (DU)

61. When a regenerative payload hosts only the gNB-DU while the CU stays on the ground, the feeder link carries the:

- A. F1 interface (F1-C signalling and F1-U user-plane traffic) ✓**
- B. Uu air interface only
- C. N6 interface toward the Internet
- D. ISL optical control channel

Answer: A. F1 interface (F1-C signalling and F1-U user-plane traffic)

62. In a multi-connectivity NTN scenario, the NTN UE:

- A. Connects to exactly one satellite at any time
- B. Uses only a wired backhaul connection
- C. Cannot connect to any terrestrial network
- D. Simultaneously communicates with multiple radio or core networks ✓**

Answer: D. Simultaneously communicates with multiple radio or core networks

63. For direct NTN access to handheld UEs within FR1, the commonly used frequency bands are:

- A. Only the mmWave FR2 bands
- B. Only optical/visible-light bands
- C. S-band and L-band ✓**
- D. Only sub-100 MHz HF bands

Answer: C. S-band and L-band

64. In the NTN HARQ retransmission process, after the receiver returns a negative acknowledgement (NACK) for a corrupted packet, the transmitter responds by:

- A. Permanently dropping the connection to the UE
- B. Retransmitting the data using a new redundancy version (RV) ✓**
- C. Switching the UE to a different satellite constellation
- D. Ignoring the NACK and sending entirely new data

Answer: B. Retransmitting the data using a new redundancy version (RV)

65. The most challenging NTN characteristic inhibiting low-latency communication is:

- A. The small physical size of satellite antennas
- B. The long round-trip time due to the large UE-to-satellite distance ✓**
- C. The absence of any modulation scheme
- D. The low cost of ground gateways

Answer: B. The long round-trip time due to the large UE-to-satellite distance

66. Typical one-way latency values quoted for NTN range from about 30-40 ms in LEO up to:

- A. About 5 ms in GEO constellations
- B. About 1 ms in GEO constellations
- C. About 544 ms in GEO constellations ✓**

D. About 60 ms in GEO constellations

Answer: C. About 544 ms in GEO constellations

67. The Doppler (carrier-frequency) shift in NTN arises mainly because:

A. The satellite is perfectly stationary in space

B. The satellite, and possibly the UE, moves, causing a time-variant carrier-frequency deviation ✓

C. The gateway periodically changes its IP address

D. The UE increases its transmit power over time

Answer: B. The satellite, and possibly the UE, moves, causing a time-variant carrier-frequency deviation

68. The 'Doppler rate' in an NTN specifically refers to:

A. The fixed frequency offset at a single instant

B. The variation of the Doppler shift over the connection time ✓

C. The data rate carried on the feeder link

D. The rate at which satellites are launched

Answer: B. The variation of the Doppler shift over the connection time

69. To compensate the uplink Doppler shift, the UE typically:

A. Estimates its position via GNSS and uses satellite ephemeris to pre-adjust its uplink carrier frequency ✓

B. Randomly hops across all available subcarriers

C. Increases transmit power to overpower the shift

D. Waits until the satellite stops moving

Answer: A. Estimates its position via GNSS and uses satellite ephemeris to pre-adjust its uplink carrier frequency

70. Ionospheric propagation that causes a rotation of the waveform's polarization is known as:

A. Doppler rotation

B. Keplerian precession

C. Rayleigh fading

D. Faraday rotation ✓

Answer: D. Faraday rotation

71. Beyond round-trip time and Doppler effects, a further NTN technical challenge affecting the PHY/MAC layers is:

A. Moving cells caused by satellite motion ✓

B. Perfectly static, unchanging coverage areas

C. The complete absence of any base station

D. An unlimited supply of available spectrum

Answer: A. Moving cells caused by satellite motion

72. In the transparent-payload protocol stack, the termination points for RRC, PDCP, RLC, MAC and PHY are on the:

A. Satellite payload and the gateway

B. NTN UE and the NTN gNB, since the satellite and gateway are transparent ✓

C. 5G core network and the Internet

D. UE and the satellite only

Answer: B. NTN UE and the NTN gNB, since the satellite and gateway are transparent

73. When a regenerative payload houses the full gNB, the RRC terminates in the satellite, which results in:

A. A much longer RRC reaction time than the transparent case

B. No RRC layer being needed at all

C. RRC termination out on the Internet

D. A shorter reaction time for RRC procedures ✓

Answer: D. A shorter reaction time for RRC procedures

74. In NTN, the gateway is the terrestrial station that:

- A. Acts as the end-user handset
- B. Generates the satellite's onboard electrical power
- C. Connects the gNB (or 5GC) functions to the satellite ✓**
- D. Provides GNSS positioning directly to the UE

Answer: C. Connects the gNB (or 5GC) functions to the satellite

75. The regenerative payload's on-board processor (OBP), inserted between the LNA and the HPA, enables:

- A. Only analogue amplification with no processing
- B. Elimination of the downlink antenna
- C. Error correction and routing of packets between beams before retransmission ✓**
- D. A direct wired connection to the 5G core

Answer: C. Error correction and routing of packets between beams before retransmission

76. At its most basic, a communication satellite acts as a:

- A. Passive mirror that reflects signals unchanged
- B. Data-storage warehouse held in orbit
- C. Repeater - a receiver linked to a transmitter using different frequencies ✓**
- D. GNSS timing clock and nothing more

Answer: C. Repeater - a receiver linked to a transmitter using different frequencies

77. A satellite communication system is divided into two main segments:

- A. The uplink segment and the billing segment
- B. The analogue segment and the digital segment
- C. The civilian segment and the military segment
- D. The space segment and the earth/ground segment ✓**

Answer: D. The space segment and the earth/ground segment

78. The communications payload of a satellite consists of two distinct parts:

- A. Solar panels and batteries
- B. Thrusters and fuel tanks
- C. Ground gateways and user terminals
- D. Repeaters/transponders and antennas ✓**

Answer: D. Repeaters/transponders and antennas

79. A transponder changes the uplink carrier frequency to a different downlink frequency. In the Ku band this is, for example, from:

- A. 14 GHz on the uplink to 11 GHz on the downlink ✓**
- B. 11 GHz on the uplink to 14 GHz on the downlink
- C. 20 GHz on the uplink to 30 GHz on the downlink
- D. 14 GHz on the uplink to 14 GHz on the downlink

Answer: A. 14 GHz on the uplink to 11 GHz on the downlink

80. If a payload bandwidth of 500 MHz is divided into channels of 36 MHz each, the approximate number of transponders is:

- A. About 12 ✓**
- B. About 500
- C. About 36
- D. About 2

Answer: A. About 12

81. A satellite repeater always uses different frequencies for receive and transmit mainly in

order to:

- A. Avoid interference between the incoming and outgoing signals ✓**
- B. Save electrical power onboard the satellite
- C. Comply with Kepler's laws of motion
- D. Reduce the satellite to a single antenna

Answer: A. Avoid interference between the incoming and outgoing signals

82. A transparent (bent-pipe) repeater typically consists of a low-noise amplifier, a high-power amplifier, and:

- A. A mixer with a local oscillator for frequency conversion ✓**
- B. An on-board baseband processor for decoding
- C. A GNSS receiver for satellite positioning
- D. A propulsion thruster for station-keeping

Answer: A. A mixer with a local oscillator for frequency conversion

83. Unlike a transparent repeater, a regenerative (processing) repeater:

- A. Simply amplifies and forwards everything, including noise
- B. Demodulates the uplink to baseband, corrects errors, then remodulates for the downlink ✓**
- C. Cannot change the carrier frequency at all
- D. Contains no amplifiers of any kind

Answer: B. Demodulates the uplink to baseband, corrects errors, then remodulates for the downlink

84. A benefit of a regenerative repeater over a bent-pipe one is that it:

- A. Consumes far less power and mass
- B. Operates without any antennas
- C. Works only with analogue signals
- D. Cleans the signal by decoding and correcting errors instead of forwarding noise ✓**

Answer: D. Cleans the signal by decoding and correcting errors instead of forwarding noise

85. Which of the following is a space-platform (bus) subsystem that supports the payload?

- A. Physical Downlink Shared Channel (PDSCH)
- B. User Plane Function (UPF)
- C. Resource Block scheduler
- D. Attitude and Orbit Control System (AOCS) ✓**

Answer: D. Attitude and Orbit Control System (AOCS)

86. The Telemetry, Tracking, and Command (TTC&M) subsystem is responsible for:

- A. Amplifying and retransmitting the user carriers
- B. Allocating IP addresses to the user devices
- C. Monitoring subsystem health, tracking position, and executing commands from ground control ✓**
- D. Performing OFDMA resource scheduling

Answer: C. Monitoring subsystem health, tracking position, and executing commands from ground control

87. The three main satellite links are the service link, the feeder link, and the:

- A. Inter-satellite link (ISL) ✓**
- B. Fibre backhaul link
- C. Ethernet control link
- D. Terrestrial microwave link

Answer: A. Inter-satellite link (ISL)

88. Inter-satellite links (ISLs) can be implemented using:

- A. Either radio-frequency or optical links ✓**
- B. Only copper cabling
- C. Only acoustic signalling

D. Only undersea fibre

Answer: A. Either radio-frequency or optical links

89. The performance of transmitting equipment is measured by its EIRP, which is:

A. The ratio of antenna gain to system noise temperature

B. The Boltzmann constant multiplied by temperature

C. The power fed to the antenna multiplied by the antenna gain in the considered direction ✓

D. The received carrier-to-noise density ratio

Answer: C. The power fed to the antenna multiplied by the antenna gain in the considered direction

90. The receiver figure of merit, G/T, is the ratio of:

A. The transmit power to the antenna gain

B. The antenna receive gain to the system noise temperature ✓

C. The carrier power to the user bit rate

D. The uplink frequency to the downlink frequency

Answer: B. The antenna receive gain to the system noise temperature

91. A satellite link budget is fundamentally used to:

A. Predict whether the received signal will be strong enough relative to the noise ✓

B. Schedule the satellite's launch window

C. Allocate telephone numbers to users

D. Determine the satellite's orbital inclination

Answer: A. Predict whether the received signal will be strong enough relative to the noise

92. Noise power spectral density N_0 is defined as:

A. The antenna gain divided by the temperature

B. The EIRP minus the free-space path loss

C. The product of the Boltzmann constant k and the system noise temperature T ✓

D. The carrier power multiplied by the bit rate

Answer: C. The product of the Boltzmann constant k and the system noise temperature T

93. Which of the following is a link (propagation) loss accounted for in a satellite link budget?

A. The Boltzmann constant of the receiver

B. The EIRP of the transmitter

C. The number of transponders onboard

D. Free-space path loss together with rain/atmospheric attenuation ✓

Answer: D. Free-space path loss together with rain/atmospheric attenuation

94. In 5G NR, Frequency Range 1 (FR1) spans approximately:

A. 410 MHz to 7.125 GHz ✓

B. 24 to 52 GHz

C. 50 to 66 GHz

D. Below 100 MHz only

Answer: A. 410 MHz to 7.125 GHz

95. In 5G NR, a resource block (RB) in the frequency domain is defined as:

A. 14 consecutive subcarriers

B. 10 consecutive subframes

C. A single OFDM symbol

D. 12 consecutive subcarriers ✓

Answer: D. 12 consecutive subcarriers

96. A 5G NR radio frame lasts 10 ms and contains:

A. One subframe of 10 ms

B. Ten subframes of 1 ms each ✓

C. Fourteen subframes of 1 ms each

D. Twelve subframes of 0.5 ms each

Answer: B. Ten subframes of 1 ms each

97. In 5G NR, the subcarrier spacing given by $\Delta f = 2^u \times 15$ kHz means that it:

A. Is fixed at 15 kHz for every deployment

B. Decreases as the numerology u increases

C. Scales flexibly with the numerology u (e.g., 15, 30, 60, 120 kHz) ✓

D. Depends only on the carrier frequency band

Answer: C. Scales flexibly with the numerology u (e.g., 15, 30, 60, 120 kHz)

98. Which set lists three 5G NR downlink physical channels?

A. PUSCH, PRACH and PUCCH

B. AMF, SMF and UPF

C. PDSCH, PDCCH and PBCH ✓

D. RRC, PDPC and RLC

Answer: C. PDSCH, PDCCH and PBCH

99. In the 5G initial access procedure, the UE first performs cell search using:

A. The random-access preamble on PRACH

B. The RRC Connection Setup message

C. The synchronization signals PSS and SSS ✓

D. The Sounding Reference Signal (SRS)

Answer: C. The synchronization signals PSS and SSS

100. To increase spectral efficiency, the 5G NR physical layer adopts which channel-coding techniques?

A. Convolutional and Reed-Solomon coding only

B. Turbo coding, exactly as used in LTE, only

C. No channel coding at all

D. Polar coding and LDPC coding ✓

Answer: D. Polar coding and LDPC coding